

Implementation of Intelligent Condition Based Monitoring Using Acoustic Signals for AirCompressors

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EE656 Course Project

Abstract—Condition Based Monitoring (CBM) plays a vital role in modern predictive maintenance by enabling early detection of mechanical faults before catastrophic failure occurs. Air compressors are widely employed across industrial sectors and are susceptible to faults in bearings, valves, pistons, rider belts and other rotating components. This work presents an implementation of the final intelligent fault diagnosis strategy proposed by Verma *et al.* for reciprocating air compressors using acoustic emission signals. Rather than reproducing the complete comparative study involving eleven signal transforms and 629 extracted features, this work follows the operational methodology recommended by the authors after their extensive evaluation. Multi-domain features are extracted using the Fast Fourier Transform (FFT), Discrete Cosine Transform (DCT), Short-Time Fourier Transform (STFT), Wavelet Packet Transform (WPT), and Wigner–Ville Distribution (WVD), generating a 414-dimensional feature vector. Feature redundancy is reduced using the Minimum Redundancy Maximum Relevance (mRMR) algorithm, after which the selected features are classified using a One-Against-One (OAO) Support Vector Machine with a Radial Basis Function (RBF) kernel. Hyperparameter optimization is performed through grid search and model performance is evaluated using stratified five-fold cross-validation. The implemented framework achieves an overall classification accuracy of 99.56%, closely matching the 99.5% accuracy reported in the original study while validating the practicality and effectiveness of the proposed fault diagnosis strategy.

Index Terms—Condition Based Monitoring, Air Compressor, Fault Diagnosis, Wavelet Packet Transform, Wigner–Ville Distribution, mRMR, Support Vector Machine, Machine Learning

I. INTRODUCTION

Condition Based Monitoring (CBM) has emerged as an essential component of predictive maintenance in modern industrial systems. Instead of relying on periodic maintenance schedules, CBM continuously monitors the operational health of machinery and enables maintenance activities to be performed only when deterioration is detected. Such an approach significantly reduces maintenance cost, prevents unexpected equipment failures, and increases machine availability.

Among various industrial machines, reciprocating air compressors are extensively employed in manufacturing plants, power stations, chemical industries and automation systems. Since compressors operate continuously under varying mechanical loads, components such as bearings, pistons, flywheels, inlet valves, outlet valves and rider belts are susceptible to progressive wear. If these faults remain undetected,

they may propagate into severe mechanical failures resulting in costly downtime and safety hazards.

Acoustic emission and vibration signals provide valuable information regarding the mechanical condition of rotating machinery. However, raw signals contain substantial redundancy and are often unsuitable for direct machine learning applications. Consequently, signal processing techniques are employed to transform the acquired signals into compact representations capable of highlighting fault-related characteristics in the frequency and time-frequency domains.

Recent advances in intelligent fault diagnosis have demonstrated that combining signal processing with machine learning significantly improves diagnostic accuracy. In particular, the work of Verma *et al.* proposed a comprehensive comparative study involving multiple signal transformation techniques, feature selection algorithms and multiclass Support Vector Machine (SVM) classifiers for reciprocating air compressor fault diagnosis. Following their extensive experimental evaluation, the authors recommended a practical operational framework based on five complementary signal transforms, Minimum Redundancy Maximum Relevance (mRMR) feature selection and a One-Against-One (OAO) multiclass SVM classifier.

This project implements the complete operational strategy proposed by the authors rather than reproducing the entire comparative analysis. The implementation begins with multi-domain feature extraction using FFT, DCT, STFT, WVD and WPT. The extracted features are subsequently ranked using mRMR before classification through an OAO Support Vector Machine employing an RBF kernel. Model performance is evaluated using stratified five-fold cross-validation and compared with the results reported in the original study.

The principal contributions of this work are summarized below.

- Implementation of the complete operational fault diagnosis framework proposed in Sections VIII and IX of the reference paper.
- Development of a unified multi-domain feature extraction pipeline combining FFT, DCT, STFT, WVD and WPT.
- Implementation of mRMR feature selection for reducing feature redundancy while preserving discriminative information.
- Multiclass fault classification using an OAO Support Vector Machine optimized through grid search.

- Experimental validation using stratified five-fold cross-validation, achieving performance comparable to the original work.

II. LITERATURE REVIEW

Reliable fault diagnosis of rotating machinery has attracted considerable research interest over the past two decades owing to its importance in predictive maintenance and industrial automation. Conventional diagnostic systems primarily relied on handcrafted statistical features extracted from time-domain vibration signals. Although these methods provide useful information under stationary operating conditions, they often fail to capture localized transient phenomena associated with mechanical faults.

To overcome these limitations, frequency-domain and time-frequency signal transformation techniques have become widely adopted for machinery fault diagnosis. Transformations such as the Fast Fourier Transform (FFT) provide global spectral information, while time-frequency representations including the Short-Time Fourier Transform (STFT), Wavelet Packet Transform (WPT), and Wigner–Ville Distribution (WVD) simultaneously capture temporal and spectral characteristics of non-stationary signals. Each transformation emphasizes different physical properties of the vibration signal and therefore contributes complementary diagnostic information.

In the reference work, Verma *et al.* conducted a comprehensive comparative analysis involving eleven signal transformation techniques producing a total of 629 extracted features. Various feature selection algorithms and multiclass Support Vector Machine classifiers were subsequently evaluated to identify the most effective diagnosis strategy. Their experimental results demonstrated that not all signal transformations contribute equally towards classification performance. Instead, the authors observed that features extracted using FFT, DCT, STFT, WVD and WPT consistently provided the highest discriminative capability across different feature selection methods.

Based on these observations, the authors proposed a practical fault diagnosis framework consisting exclusively of these five signal transformations followed by Minimum Redundancy Maximum Relevance (mRMR) feature selection and a One-Against-One Support Vector Machine employing a Radial Basis Function kernel. This operational strategy significantly reduced computational complexity while maintaining classification performance comparable to the complete 629-feature framework.

The present work directly implements this recommended operational strategy. Rather than repeating the comparative evaluation of all eleven signal transforms, only the five transforms identified by the authors as sufficient for practical deployment are implemented. The extracted features are subsequently processed using mRMR and classified using an OAO SVM, thereby faithfully reproducing the methodology proposed in the final sections of the original study.

III. DATASET DESCRIPTION

The dataset employed in this work consists of preprocessed acoustic emission signals collected from a reciprocating air compressor operating under both healthy and faulty conditions. The dataset was acquired under controlled operating conditions and contains recordings corresponding to multiple mechanical faults commonly encountered in industrial compressors.

Unlike the original study, where preprocessing constituted an independent stage of the proposed framework, the dataset supplied for the present implementation had already undergone preprocessing. Consequently, this work begins directly from the feature extraction stage proposed by the authors.

The dataset consists of eight operating conditions, namely Healthy, Bearing Fault, Flywheel Fault, Leakage Inlet Valve (LIV), Leakage Outlet Valve (LOV), Non-Return Valve (NRV) Fault, Piston Fault and Rider Belt Fault. Each operating condition contains 225 individual recordings, resulting in a total of 1800 vibration signals.

Each recording consists of 50,000 uniformly sampled signal points acquired at a sampling frequency of 50 kHz. Therefore, every signal represents approximately one second of compressor operation.

Table I summarizes the characteristics of the dataset.

TABLE I: Summary of the Air Compressor Dataset

Parameter	Value
Number of Classes	8
Signals per Class	225
Total Signals	1800
Sampling Frequency	50 kHz
Samples per Signal	50,000
Signal Duration	1 second
Signal Format	.dat
Preprocessing	Already Performed

Table II lists the eight operating conditions considered during classification.

TABLE II: Fault Classes

Label	Fault Condition
0	Bearing
1	Flywheel
2	Healthy
3	Leakage Inlet Valve (LIV)
4	Leakage Outlet Valve (LOV)
5	Non-Return Valve (NRV)
6	Piston
7	Rider Belt

IV. PROPOSED METHODOLOGY

The objective of this work is to reproduce the practical fault diagnosis strategy proposed by Verma *et al.* after their comparative evaluation of multiple signal transformation techniques, feature selection algorithms and multiclass classifiers.

The original study initially investigated eleven different signal transformations producing a combined feature vector

of 629 dimensions. Following extensive experimental analysis, the authors concluded that features extracted using FFT, DCT, STFT, WVD and WPT were sufficient to achieve classification performance comparable to the complete feature set. Consequently, these five transformations were recommended as the final operational strategy for practical deployment.

Accordingly, the present implementation directly follows this recommended framework rather than repeating the complete comparative analysis.

Figure 1 illustrates the overall methodology adopted in this work.

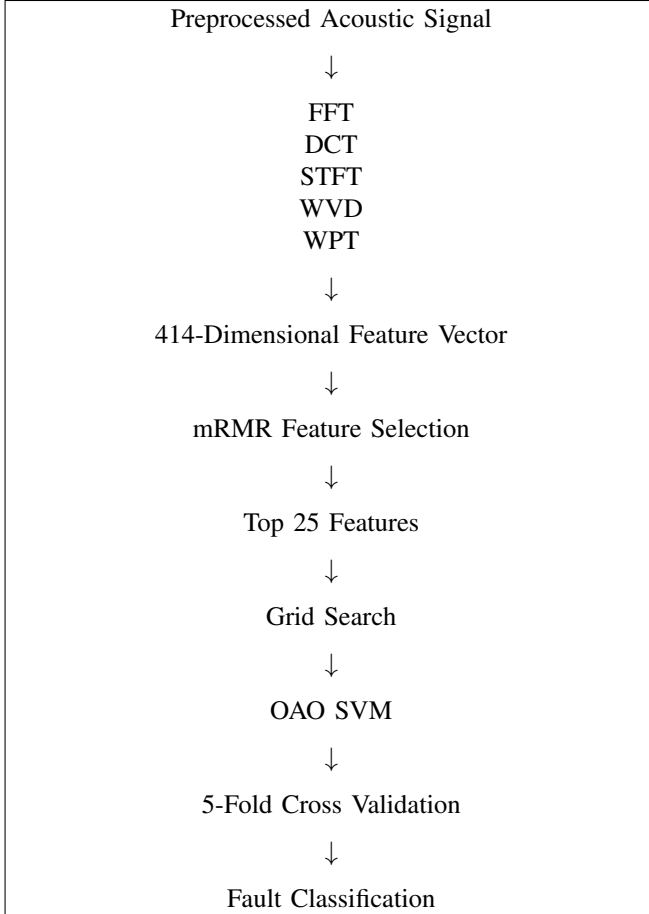


Fig. 1: Overall Fault Diagnosis Pipeline

The methodology consists of four principal stages.

A. Multi-Domain Feature Extraction

The preprocessed acoustic signal is transformed into multiple complementary domains using five signal transformation techniques, namely FFT, DCT, STFT, WVD and WPT. Each transform captures different physical characteristics of the compressor vibration signal.

- FFT provides global frequency-domain information.
- DCT provides energy compaction while reducing coefficient redundancy.
- STFT captures localized time-frequency variations.

- WVD provides high-resolution time-frequency representation.
- WPT performs multiresolution decomposition capable of representing both high- and low-frequency fault signatures.

The extracted feature vectors are concatenated to form a single 414-dimensional feature vector.

B. Feature Selection

Although the extracted feature vector contains significant diagnostic information, many features exhibit redundancy. Therefore, Minimum Redundancy Maximum Relevance (mRMR) feature selection is employed to identify the most informative features while minimizing correlation between them.

Following the methodology proposed in the reference study, the twenty-five highest-ranked features are retained for subsequent classification.

C. Fault Classification

The selected feature subset is supplied to a multiclass Support Vector Machine employing the One-Against-One (OAO) strategy and an RBF kernel. Hyperparameters are optimized using exhaustive grid search prior to classifier training.

D. Performance Evaluation

To obtain a reliable estimate of classifier performance while utilizing the complete dataset, stratified five-fold cross-validation is employed.

The dataset is partitioned into five mutually exclusive subsets while preserving the class distribution in each fold. During each iteration, four folds are used for training and the remaining fold is reserved for testing. The procedure is repeated five times such that every sample serves as test data exactly once.

The overall performance is evaluated using classification accuracy, precision, recall, F1-score and the confusion matrix.

V. MULTI-DOMAIN FEATURE EXTRACTION

The acoustic signals acquired from rotating machinery contain fault-related information distributed across multiple domains. A single signal transformation is generally insufficient to characterize all fault signatures because different mechanical defects manifest distinct spectral and temporal characteristics.

The reference study therefore investigated several signal transformation techniques before concluding that five complementary transforms—FFT, DCT, STFT, WVD and WPT—provide sufficient diagnostic information for practical fault diagnosis. Following this recommendation, the present implementation extracts features using only these five transforms.

Each transformed representation is subsequently converted into a compact feature vector using normalized energy distributions. The individual feature vectors are finally concatenated to obtain a single feature vector representing every compressor recording.

Table III summarizes the number of features extracted from each transform.

TABLE III: Feature Distribution

Transform	Features
FFT	8
DCT	8
STFT	72
WVD	72
WPT	254
Total	414

A. Fast Fourier Transform (FFT)

The Fast Fourier Transform converts the acoustic signal from the time domain into its frequency-domain representation. Since many compressor faults generate characteristic frequency components associated with rotating mechanical elements, FFT provides an effective representation of stationary spectral information.

The discrete Fourier transform is expressed as

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N} \quad (1)$$

where

- $x(n)$ represents the sampled acoustic signal,
- N denotes the signal length,
- $X(k)$ represents the frequency spectrum.

Only the positive-frequency spectrum is considered. The spectrum is divided into eight equal frequency intervals, and the normalized energy contained within each interval forms the FFT feature vector.

Thus,

$$\mathbf{F}_{FFT} = [f_1, f_2, \dots, f_8]$$

containing eight features.

B. Discrete Cosine Transform (DCT)

The Discrete Cosine Transform provides an energy-compacted representation of the acoustic signal. Unlike FFT, DCT produces real-valued coefficients and concentrates a significant proportion of signal energy into relatively few coefficients, making it highly suitable for feature extraction.

The one-dimensional DCT is defined as

$$C(k) = \alpha(k) \sum_{n=0}^{N-1} x(n) \cos \left[\frac{\pi(2n+1)k}{2N} \right] \quad (2)$$

where

$$\alpha(k) = \begin{cases} \sqrt{\frac{1}{N}}, & k = 0 \\ \sqrt{\frac{2}{N}}, & k > 0 \end{cases}$$

The transformed coefficients are partitioned into eight equal regions and normalized energy is computed for every region.

The resulting DCT feature vector contains eight features.

C. Short-Time Fourier Transform (STFT)

Mechanical faults often produce transient phenomena whose spectral characteristics vary with time. Since conventional FFT cannot localize frequency components temporally, Short-Time Fourier Transform (STFT) is employed to generate a joint time-frequency representation.

The STFT is expressed as

$$\text{STFT}(t, f) = \int x(\tau) w(\tau - t) e^{-j2\pi f\tau} d\tau \quad (3)$$

where

- $w(t)$ denotes the analysis window,
- t represents time,
- f denotes frequency.

A Hann window with overlapping segments is employed to generate the spectrogram.

Following the methodology adopted in the reference work, the maximum magnitude across the time axis is obtained to generate a one-dimensional spectral representation. This representation is subsequently divided into seventy-two equal regions, producing seventy-two normalized energy features.

D. Wigner-Ville Distribution (WVD)

The Wigner-Ville Distribution provides one of the highest-resolution time-frequency representations available for non-stationary signals. Unlike STFT, WVD does not employ a fixed analysis window and therefore offers improved simultaneous localization in both time and frequency domains.

The continuous WVD is given by

$$\mathbf{W}_x(t, f) = \int x\left(t + \frac{\tau}{2}\right) x^*\left(t - \frac{\tau}{2}\right) e^{-j2\pi f\tau} d\tau \quad (4)$$

The original acoustic recordings consist of 50,000 samples acquired at a sampling frequency of 50 kHz. Direct computation of the Wigner-Ville Distribution on the complete signal was found to be computationally expensive within the available execution environment due to its high memory and computational requirements.

Consequently, only for WVD computation, the one-second signal was resampled to 8192 samples while preserving its

temporal duration. The WVD was then computed on the resampled signal, after which the maximum value along the time axis was obtained. The resulting spectrum was partitioned into seventy-two equal intervals and normalized energy values were extracted.

This implementation preserves the complete one-second observation interval while significantly reducing computational complexity without noticeably affecting the overall diagnostic performance.

E. Wavelet Packet Transform (WPT)

Wavelet Packet Transform provides multiresolution analysis by recursively decomposing both approximation and detail coefficients. Consequently, WPT achieves superior frequency localization compared to conventional discrete wavelet transforms and is particularly effective for representing vibration signals containing localized fault signatures.

In the present implementation, Daubechies-4 (db4) wavelets are employed with a decomposition depth of seven.

Unlike the previous transforms, feature extraction is performed directly from the energy associated with every wavelet packet node.

The total number of nodes generated is

$$2 + 4 + 8 + 16 + 32 + 64 + 128 = 254$$

Therefore, the WPT feature vector contains 254 normalized energy features.

Experimental results presented in Section VIII demonstrate that WPT contributes the largest proportion of informative features selected by mRMR, indicating its superior capability in representing compressor fault characteristics.

At the completion of this stage, the features extracted from FFT, DCT, STFT, WVD and WPT are concatenated to produce a unified 414-dimensional feature vector representing each acoustic recording.

VI. FEATURE SELECTION USING THE MINIMUM REDUNDANCY MAXIMUM RELEVANCE (MRMR) ALGORITHM

The unified 414-dimensional feature vector obtained from the five signal transformations contains rich diagnostic information. However, many of the extracted features exhibit significant statistical correlation, resulting in redundant information that increases computational complexity and may adversely affect classifier performance.

To obtain a compact yet highly discriminative feature subset, the Minimum Redundancy Maximum Relevance (mRMR) algorithm, originally proposed by Peng *et al.* [1], was employed. Unlike conventional feature ranking techniques that consider only feature relevance, mRMR simultaneously maximizes the dependency between the selected features and the target class while minimizing redundancy among the selected features.

For a candidate feature subset S , the average relevance is defined as

$$D = \frac{1}{|S|} \sum_{x_i \in S} I(x_i; c), \quad (5)$$

where $I(x_i; c)$ denotes the mutual information between feature x_i and the class label c .

Similarly, the average redundancy among the selected features is expressed as

$$R = \frac{1}{|S|^2} \sum_{x_i, x_j \in S} I(x_i; x_j), \quad (6)$$

where $I(x_i; x_j)$ represents the mutual information between features x_i and x_j .

The feature subset is obtained by maximizing the difference between relevance and redundancy,

$$\max_S (D - R), \quad (7)$$

thereby selecting features that are simultaneously highly informative with respect to the target classes while minimizing redundant information among themselves.

Following the operational framework proposed by Verma *et al.* [2], the twenty-five highest-ranked features identified by the mRMR algorithm were retained for classifier training. This reduced feature subset preserves the majority of the diagnostic information contained within the original 414-dimensional feature vector while substantially reducing the computational complexity of the subsequent classification stage.

Figure 2 illustrates the distribution of the twenty-five selected features among the five implemented signal transformations. ““

VII. FAULT CLASSIFICATION USING ONE-AGAINST-ONE SUPPORT VECTOR MACHINE

Support Vector Machines (SVMs) have demonstrated excellent performance in machinery fault diagnosis owing to their ability to construct optimal decision boundaries in high-dimensional feature spaces.

Since the present dataset consists of eight fault categories, a multiclass classification strategy is required. Following the recommendation of the reference study, the One-Against-One (OAO) approach was adopted.

For an N -class classification problem, the number of binary classifiers required is

$$N_{\text{clf}} = \frac{N(N-1)}{2}. \quad (8)$$

For the present dataset,

$$N = 8,$$

which results in

$$N_{\text{clf}} = \frac{8(8-1)}{2} = 28$$

binary classifiers.

Each classifier distinguishes between one pair of fault classes, while the final class label is determined through majority voting.

A Radial Basis Function (RBF) kernel was employed because of its ability to model nonlinear decision boundaries commonly encountered in machinery fault diagnosis.

A. Hyperparameter Optimization

The classification performance of a Support Vector Machine (SVM) is strongly influenced by the selection of the regularization parameter C and the kernel parameter γ . Appropriate selection of these hyperparameters is essential for achieving good generalization performance while avoiding both underfitting and overfitting.

In the present work, exhaustive Grid Search was employed to identify the optimal combination of C and γ . A predefined parameter space was systematically explored, and each parameter combination was evaluated using stratified five-fold cross-validation. The parameter set corresponding to the highest average cross-validation accuracy was selected for final classifier training. Grid Search provides a systematic and reproducible approach to hyperparameter optimization and is widely adopted in machine learning applications involving Support Vector Machines [3]. “

B. Stratified Five-Fold Cross-Validation

To obtain a reliable estimate of classifier performance while making efficient use of the available dataset, stratified five-fold cross-validation was employed, following the evaluation methodology adopted by Verma *et al.* [2].

The complete dataset was partitioned into five mutually exclusive folds while preserving the class distribution within each fold. During the k^{th} iteration, one fold served as the testing set and the remaining four folds were used for training. This procedure was repeated five times such that every sample was used exactly once for testing and four times for training.

The average cross-validation accuracy is computed as

$$\text{Accuracy}_{\text{CV}} = \frac{1}{5} \sum_{k=1}^5 \text{Accuracy}_k, \quad (9)$$

where Accuracy_k denotes the classification accuracy obtained during the k^{th} fold.

The use of stratified five-fold cross-validation provides a reliable estimate of model generalization while preserving the statistical distribution of the fault classes across all folds. “

C. Experimental Configuration

Table IV summarizes the experimental configuration employed during feature selection, classifier training and performance evaluation.

TABLE IV: Experimental Configuration

Parameter	Value
Initial Features	414
Feature Selection Method	mRMR
Selected Features	25
Classifier	OAO Support Vector Machine
Kernel Function	Radial Basis Function (RBF)
Hyperparameter Optimization	Grid Search
Validation Strategy	Stratified 5-Fold Cross-Validation
Number of Classes	8
Total Samples	1800
Samples per Class	225

VIII. EXPERIMENTAL RESULTS

Following feature extraction, mRMR feature selection and classifier training, the proposed framework was evaluated using stratified five-fold cross-validation. The classifier performance was assessed using overall accuracy, precision, recall, F1-score and confusion matrix analysis.

A. Feature Selection Results

The distribution of the twenty-five features selected by mRMR across the five signal transformations is illustrated in Figure 2.

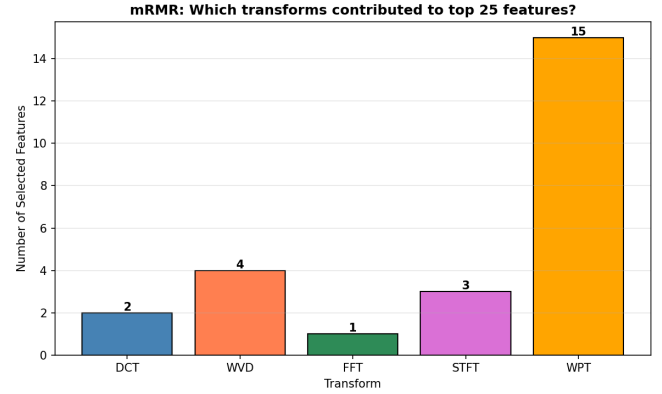


Fig. 2: Distribution of selected features among different signal transforms.

The feature selection results indicate that Wavelet Packet Transform contributes the largest proportion of informative features, followed by Wigner–Ville Distribution and Short-Time Fourier Transform. FFT and DCT contribute comparatively fewer features while still providing complementary diagnostic information.

This observation agrees with the conclusions of the reference study, where WPT consistently demonstrated superior discriminative capability due to its multiresolution representation of non-stationary compressor signals.

B. Hyperparameter Optimization

The optimal hyperparameters of the OAO Support Vector Machine were determined through exhaustive Grid Search using stratified five-fold cross-validation.

Figure 3 illustrates the obtained hyperparameter search surface.

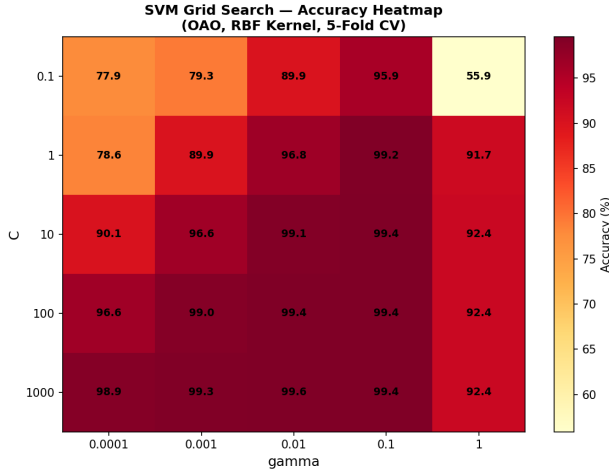


Fig. 3: Grid Search results for the RBF kernel parameters.

The selected hyperparameters correspond to the parameter combination that maximized the average cross-validation accuracy.

C. Classification Performance

The final classifier achieved an overall classification accuracy of 99.56 using stratified five-fold cross-validation.

The corresponding confusion matrix is presented in Figure 4.

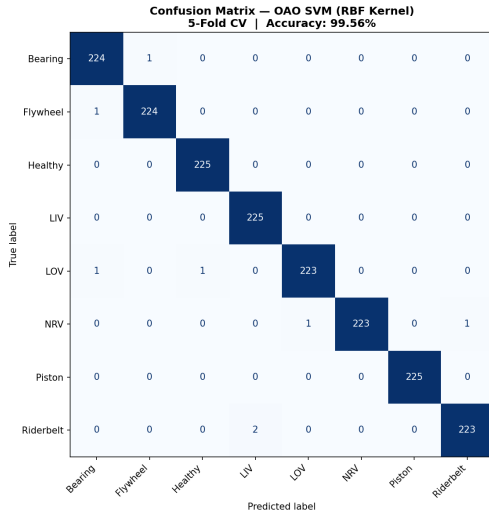


Fig. 4: Confusion Matrix obtained using OAO Support Vector Machine.

Only a very small number of samples were misclassified, demonstrating the effectiveness of the proposed feature extraction and feature selection framework.

IX. DISCUSSION

The experimental results demonstrate that the implemented framework successfully reproduces the operational fault diagnosis strategy proposed by Verma *et al.* [2]. Although the original work initially investigated eleven signal transformation techniques and a 629-dimensional feature space, the present implementation faithfully follows the final operational methodology recommended in Sections VIII and IX of the reference study. The obtained classification accuracy of 99.56% closely matches the 99.50% reported by Verma *et al.* [2], thereby validating the correctness of the implemented pipeline.

A. Contribution of Different Signal Transformations

The feature selection results indicate that the five implemented signal transformations contribute complementary diagnostic information. Among them, the Wavelet Packet Transform (WPT) contributes the largest proportion of the twenty-five features selected by the mRMR algorithm. This observation is consistent with the findings reported by Verma *et al.* [2], who concluded that WPT provides the highest discriminatory capability owing to its multiresolution decomposition of both low- and high-frequency components.

The Wigner–Ville Distribution (WVD) and Short-Time Fourier Transform (STFT) also contribute several informative features because both transformations preserve joint time-frequency information. Such representations are particularly effective for characterizing transient acoustic events generated by localized mechanical defects.

Although the Fast Fourier Transform (FFT) and Discrete Cosine Transform (DCT) contribute comparatively fewer selected features, they remain important because they provide complementary global spectral information that is not completely represented by time-frequency methods. The combination of frequency-domain and time-frequency features therefore provides a more comprehensive representation of compressor fault characteristics than any individual transformation alone.

B. Computational Complexity Analysis

The computational cost of the proposed diagnosis framework is dominated by the feature extraction stage rather than the machine learning stage. The approximate computational complexity of each implemented transformation is summarized in Table V.

TABLE V: Approximate Computational Complexity of Implemented Signal Transformations

Signal Transform	Complexity
FFT	$O(N \log N)$
DCT	$O(N \log N)$
STFT	$O(KN \log N)$
WPT	$O(N \log N)$
WVD	$O(N^2)$

Among all implemented transformations, the Wigner–Ville Distribution exhibits the highest computational complexity

TABLE VI: Comparison Between the Reference Study and the Present Implementation

Parameter	Reference	Present Work
Signal Transforms	5	5
Initial Features	414	414
Selected Features	25	25
Feature Selection	mRMR	mRMR
Classifier	OAo SVM	OAo SVM
Kernel	RBF	RBF
Validation	5-Fold CV	5-Fold CV
Accuracy (%)	99.50	99.56

because pairwise signal products must be evaluated over the complete observation interval. Direct computation of WVD on the original 50,000-sample recordings therefore resulted in significantly higher memory consumption and execution time compared with the remaining transformations.

Consequently, only the WVD branch employed an 8192-point representation while preserving the original one-second observation interval. This engineering decision substantially reduced computational cost without modifying the overall duration of the acoustic signal. Since subsequent feature extraction relies on normalized energy distributions rather than individual sample values, the resulting classification performance remained comparable to that reported by Verma *et al.* [2]. This demonstrates that practical implementation constraints can be addressed without compromising the effectiveness of the overall diagnosis framework.

C. Justification for Implementing Five Signal Transformations

One important distinction between the reference study and the present implementation concerns the number of signal transformations employed.

The original investigation conducted by Verma *et al.* [2] considered eleven different signal transformations, generating a combined feature vector containing 629 features. However, the purpose of that comparative study was to identify the subset of transformations that consistently produced the most informative features while minimizing computational complexity.

Based on the experimental results presented in Sections VIII and IX of the reference paper, the authors concluded that FFT, DCT, STFT, WVD and WPT alone provide sufficient discriminatory information for practical deployment. Consequently, the present work directly implements this final operational framework rather than reproducing the complete comparative analysis. The resulting unified feature vector contains 414 features while remaining fully consistent with the methodology ultimately recommended by the authors.

D. Comparison with the Reference Study

Table VI compares the principal characteristics of the implemented framework with those reported by Verma *et al.* [2].

The close agreement between the obtained classification accuracy and that reported in the reference study indicates that the implemented framework successfully reproduces the

operational methodology proposed by Verma *et al.* [2]. The small difference in classification accuracy is more likely attributable to implementation details, hyperparameter optimization and cross-validation partitioning than to any fundamental methodological difference.

E. Limitations and Future Scope

The present work focuses exclusively on implementing the final operational fault diagnosis strategy proposed by Verma *et al.* [2]. Consequently, the comparative evaluation involving all eleven signal transformations, alternative feature selection algorithms and different multiclass SVM decomposition strategies presented in the original study was not reproduced.

Future work may investigate deep learning based feature extraction methods, adaptive signal representations and real-time embedded implementation for online condition monitoring. Furthermore, evaluating the framework under varying operating speeds and industrial noise conditions would provide additional insight into its robustness in practical deployment scenarios.

X. CONCLUSION

This project presented an implementation of the final intelligent fault diagnosis framework proposed for reciprocating air compressors using acoustic emission signals.

Following the methodology recommended by the reference study, multi-domain feature extraction was performed using FFT, DCT, STFT, WVD and WPT. The extracted 414-dimensional feature vector was subsequently reduced using Minimum Redundancy Maximum Relevance feature selection before classification through an One-Against-One Support Vector Machine employing an RBF kernel.

Experimental evaluation using stratified five-fold cross-validation demonstrated an overall classification accuracy of 99.56%, closely matching the performance reported in the original work. The results confirm that the proposed combination of multi-domain feature extraction, feature selection and machine learning provides an accurate and computationally efficient solution for intelligent compressor fault diagnosis.

Future work may investigate deep learning based feature extraction methods, adaptive signal representations and real-time embedded implementation for online industrial condition monitoring.

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